DataWeb - Problem Statements

Time Duration : 6 hours

Track 1 : GenAI Track

PS 1 : Investigation System for UPI and Cheque Transactions

Banks process large volumes of transactions through UPI and cheque payments. Automated monitoring systems flag certain transactions that show unusual patterns or inconsistencies.

These flagged cases enter an **Exception Investigation Phase**, where operations officers analyze:

- Customer account details
- Transaction history
- Merchant information
- Device and location data
- Cheque details (if applicable)

Due to increasing transaction volumes, banks require intelligent systems that can assist officers by analyzing data and generating explainable investigation outcomes.

Objective

Design and develop an **Intelligent Investigation Assistant using Generative AI** that:

- Analyzes flagged UPI transactions (no need of implementing flagging mechanism)
- (Optional) Investigates cheque payment cases
- Detects inconsistencies and anomalies

- Generates explainable investigation outcomes

The system should simulate a real-world banking investigation workflow.

Investigation Scope

- 1. UPI Transaction Investigation
- 2. Cheque Investigation (Good to Have, Optional Use Case)

Required Capabilities

The system must be capable of:

- Analyzing customer, transaction, and merchant data
- Correlating UPI transaction data across datasets
- Extracting information from cheque images (if implemented)
- Identifying inconsistencies between payment data and customer records
- Using Generative AI for reasoning and report generation

Generative AI must be a **core component** of investigation reasoning and structured report generation

Required Output

For each flagged case, the system must generate:

1. Investigation Outcome

2. Investigation Report

A structured explanation that clearly states:

- Data analyzed
- Detected inconsistencies
- Reasoning behind the final decision

Utilising GenAI Capabilities for the same

Dataset Generation Guidelines

Refer to the dataset structure and schema here:

https://gist.github.com/DYNAMICMORTAL/b2df80b5df5211215ccd6b6714a87157

Participants may generate sample data following the provided schema and relationships.

Goal

Build an intelligent investigation system that analyzes structured financial data and cheque instruments to produce accurate, explainable investigation decisions using Generative AI.

PS 2 : Conversational Data Intelligence Platform

Modern businesses generate large volumes of structured data, but non-technical stakeholders often struggle to extract insights without relying on data teams.

There is a growing need for reliable, explainable, and hallucination-free AI systems that allow business users to query data using natural language and receive accurate, data-backed answers.

Objective

Design and develop a **Conversational Data Intelligence Platform** that:

- Enables users to ask questions in natural language
- Analyzes structured datasets
- Generates accurate, data-backed responses
- Provides explainable outputs

The system must ensure that all answers are grounded strictly in the underlying dataset.

Scope

The system should:

- Ingest structured CSV datasets
- Understand user queries in natural language
- Retrieve relevant data from the dataset
- Generate accurate responses based only on actual data

The solution must be **dataset-agnostic**, meaning it should work with any structured dataset following a defined schema.

Required Capabilities

The system must:

- Process and structure raw CSV data
- Interpret business questions in natural language
- Retrieve and analyze relevant data
- Generate responses strictly grounded in database results
- Avoid hallucinations
- Provide transparency in how the answer was derived
- Generate visualizations when appropriate graphs where relevant (optional, good to have)

Required Output (Per Query)

For each user query, the system must return:

1. **Data-Backed Natural Language Answer**
2. **The Executed Query or Retrieval Logic Used**
3. **Brief Explanation of How the Answer Was Derived**
4. **Relevant Visualization (if applicable, good to have)**

All responses must strictly reflect actual dataset values.

Example Query

"What is the Churn Rate Analysis of Male and Female Customers"

Dataset Guidelines

The system must be dataset-agnostic and work with structured CSV files.

Sample dataset :
https://github.com/Geo-y20/Telco-Customer-Churn-/blob/main/Telco%20Customer%20Churn.csv

Participants may use any structured dataset for demonstration, provided the system can adapt to schema changes.

Goal

Build a reliable conversational analytics system that enables business users to extract accurate, explainable insights from structured data using natural language.

Track 2 : Data & Automation

PS 3 : Real-Time Aviation Intelligence Pipeline

The aviation industry generates continuous streams of operational data including flight movements, delays, geographic distribution, and airline activity. However, transforming raw API data into structured, actionable intelligence requires robust data engineering practices.

In this challenge, participants must build an end-to-end automated data pipeline that ingests live aviation data from a publicly available API and converts it into meaningful real-time analytics.

Website: https://docs.apilayer.com/aviationstack/docs/api-documentation

The solution must:

1. Fetch real-time or near real-time aviation data from the API
2. Perform cleaning and transformation through an ETL workflow
3. Store processed data in a structured database
4. Present insights via a dashboard or web-based analytics interface

The dashboard should include metrics such as:

- Active flights
- Airline distribution
- Delay patterns
- Geographic visualization of flight activity

Automation tools such as Airflow, n8n, Kestra, or scheduled workflows may be used. The focus is on pipeline architecture, automation, data modeling, and real-time insight generation, not on building complex UI systems.

PS 4 : Global Air Quality & Weather Intelligence System

Air pollution is a major global environmental challenge. Understanding air quality trends and their relationship with weather conditions can help governments and agencies take timely action.

Participants must build a **data engineering and automation pipeline** that integrates real-time air quality data with weather data to generate actionable environmental insights.

Objective

Design and develop a system that:

- Collects real-time air quality and weather data
- Cleans and standardizes the data
- Integrates both datasets
- Generates analytical insights
- Provides automated alerts based on thresholds

Focus should be on **data ingestion, transformation, automation, and analytical clarity**, not complex predictive modeling.

Data Sources

- WAQI API (World Air Quality Index) [You can use your own)
  https://aqicn.org/data-platform/token/
- OpenWeatherMap API

Scope

The system must:

- Ingest air quality and weather data from APIs
- Clean and normalize datasets
- Store structured data)
- Automate periodic data fetching
- Provide analytical dashboard

Required Capabilities

The solution should include:

- Automated data ingestion pipeline
- Data transformation and integration logic
- Structured storageDashboard for visualization
- Configurable AQI threshold detection
- Automated alert system

Required Output

1. Analytical Dashboard

The dashboard must show:

- AQI trends over time
- City-wise AQI comparison
- Correlation between weather conditions (temperature, humidity, wind, etc.) and pollution levels
- Visual summaries suitable for decision-makers

2. Automated Alerts

The system must trigger alerts when:

- AQI crosses defined danger thresholds
- Sudden spikes occur
- Hazardous pollution levels are detected

Alerts may include:

- SMS pings (simulated or real)
- Email notifications
- Console-based alert logs

Expected Outcomes

A strong submission should demonstrate:

- Working automated data ingestion
- Clean and merged air quality + weather dataset
- Meaningful visual dashboard
- Threshold-based alert mechanism
- Clear automation workflow

Goal

Build an automated environmental pipeline that integrates air quality and weather data to provide real-time insights and proactive alerting for decision-makers.

Automation tools such as Airflow, n8n, Kestra, or scheduled workflows may be used. The focus is on pipeline architecture, automation, data modeling, and real-time insight generation.